MULTI-CRITERIA OPTIMIZATION OF ELECTRIC VEHICLE CHARGING AND ROUTE PLANNING

TERM REPORT

CENG415 SENIOR DESIGN PROJECT AND SEMINAR 1
(Academic)

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DECEMBER 2025

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ABSTRACT

The transition to sustainable transportation is currently hampered by range anxiety and the unpredictability of charging infrastructure particularly for individual Electric Vehicle (EV) drivers. While existing literature and commercial applications offer sophisticated routing solutions, they predominantly focus on fleet logistics and single parameter focused optimization like financial cost, often neglecting the specific needs of the common driver. This project addresses this critical gap by developing a multi-criteria optimization engine tailored specifically for the individual EV user.

To solve this Multi-Objective Electric Vehicle Routing Problem (EVRP), the project uses a hybrid matheuristic framework that integrates the exactness of Mixed Integer Linear Programming (MILP) with the computational speed of Adaptive Large Neighborhood Search (ALNS). The problem is mathematically modeled as a resource-constrained shortest path problem on a directed graph, where the objective function is a weighted sum of time, cost, and a battery-state penalty with inclusion of external variables like weather and road conditions.

The project execution follows a structured timeline divided into seven work packages that is planned to be completed in an 8-month period.

This project meets the United Nations Sustainable Development Goals (SDGs), specifically contributing to sustainability goals. By democratizing access to advanced routing optimization and encourages wider EV adoption. The economic impact is realized through direct cost savings for users via energy efficient routing, while the reduction of range anxiety directly improves the mental well-being and safety of drivers. Ultimately, the project aims to accelerate the shift towards green mobility by making electric vehicle travel as stress-free as conventional driving.

Keywords: Electric Vehicle Routing, Optimization, MILP, ALNS

1. RESEARCH CONTRIBUTION

1.1. Significance of the Study, Contribution of the Project and Research Questions

The mass adoption of EVs by individual users is still a major question mark because of the fear of depleting the battery before reaching a destination, and the uncertainty regarding the availability and functionality of charging stations. While EVs offer lower running costs and modern technology, current navigation solutions often fail to address the complex, multi-faceted nature of EV travel for the common user.

Most existing commercial navigation applications and academic studies focus on logistics and fleet management such as optimizing for total fleet distance or aggregate delivery times rather than the specific needs of an individual driver. Most of these studies highly focus on issues that are not beneficial to the individual user, such as returning to starting point, heterogeneous vehicle fleet management, and time spent at the customer's premises. Meanwhile current individual-use applications often restrict users to specific operator networks[14], prioritize the shortest path without considering critical variables such as charging costs, station wait times due to congestion of usage[5][6], and the impact of external factors (weather, traffic) on battery efficiency and time constraints[7][15][16] or utilize static averages rather than using dynamic data[17]. This project addresses these shortcomings by developing a multi-criteria optimization framework specifically designed for individual EV users, moving beyond simple distance minimization to a holistic evaluation of travel time, total financial cost, and charging reliability.

Q1: How can a multi-criteria optimization framework be effectively formulated to balance conflicting individual user objectives specifically travel time, charging cost, and range anxiety[10]?

Most existing literature formulates the EVRP objective function with a focus on logistics costs. For instance, Abid et al. (2024)[9] proposes a multi-objective function that minimizes total travel time, distance, energy consumed, and the number of vehicles deployed. Similarly, Xia et al. (2024)[3] utilizes a "fitness function" within a two-stage heuristic to evaluate solutions based on distance, capacity compliance, and electric feasibility.

Q2: How does the inclusion of real-world dynamic constraints (capacitated station availability, weather-impacted battery discharge, and traffic etc.) improve the reliability and efficiency of EVRP compared to standard shortest-path algorithms?

The impact of physical and environmental constraints is well-documented in recent studies. Bruglieri et al. (2025)[2] address the "Capacitated Recharging Station (CRS)" problem, demonstrating that accounting for limited plug availability prevents infeasible routes where vehicles arrive at full stations, a critical reliability factor. Furthermore, Abid et al. (2024)[9] explicitly model the impact of weather (temperature), road slope, and traffic speed on energy consumption, proving that ignoring these factors leads to unrealistic range estimations and potential battery depletion.

Q3: Can a hybrid algorithmic approach combining MILP and ALNS provide real-time, optimal routing recommendations that are computationally feasible for a multi-criteria optimization engine?

The efficiency of hybrid Matheuristics for complex EVRP variants is strongly supported by Bruglieri et al. (2025)[2] and Ferone et al. (2025)[8], who developed an ALNS metaheuristic that utilizes a MILP sub-problem to check feasibility and schedule charging. Their results show that this hybrid approach outperforms exact methods (like Branch-and-Cut) in large-scale instances regarding computational time and solution quality. Additionally, Wang et al. (2025)[4] validate the use of ALNS and Variable Neighborhood Search (VNS) for handling complex, multi-attribute constraints (like heterogeneous fleets and multiple trips), confirming ALNS's flexibility and speed.

By answering these questions, the project aims to bring a new point of view to the EVRP compared to complex academic routing models and imply practical, everyday EV navigation, directly contributing to more efficient and sustainable personal transportation.

1.2. Research Objectives

The primary goal of this project is to develop a comprehensive, user-centric route planning system that mitigates range anxiety and optimizes travel for individual electric vehicle owners.

The project will develop a Mixed Integer Linear Programming (MILP) formulation that mathematically defines the trade-offs between minimizing travel time, total financial cost, and energy consumption. This model will explicitly incorporate advanced constraints found in the literature, specifically partial recharging strategies (allowing flexible charging amounts rather than mandatory full charges) and capacitated station logic (accounting for limited plug availability), adapted from fleet logistics to individual use.

One of the main focuses will be the design and implementation of a hybrid algorithmic framework that utilizes MILP for exact solutions on small scale instances and an Adaptive Large Neighborhood Search (ALNS) metaheuristic for large-scale, real-time routing. The objective is to achieve high quality solutions with low computational variance, ensuring the application can respond to user queries rapidly.

In the implementation, a data-driven fitness function that moves beyond simple distance calculation by integrating dynamic APIs. This involves processing traffic data (TomTom)[11], charging station related data (EPDK Şarj@TR)[13], and information on weather condition (Open-Meteo)[12] to provide even more realistic range and time estimations.

For usage quality, a Python-based prototype will be implemented to visualize the optimized routes and charging schedules. This objective includes validating the system's effectiveness by comparing the proposed multi-criteria routes against standard shortest-path benchmarks to demonstrate improvements in cost efficiency and travel reliability.

2. METHODOLOGY

The methodology for this project adopts a simulation-based approach, integrating mathematical modeling with algorithmic engineering. At the heart of the project lies an optimization engine built with this engineering approach.

To address the computational complexity of the Electric Vehicle Routing Problem (EVRP) while ensuring the solution is viable for individual use, the study employs a Matheuristic Framework. This hybrid framework combines the exactness of Mixed Integer Linear Programming (MILP) with the speed of Adaptive Large Neighborhood Search (ALNS), a technique validated in recent literature for solving complex routing constraints like partial recharging and capacitated stations.

2.1. Mathematical Formulation (MILP)

The problem is modeled as a Multi-Objective EVRP on a directed graph $G=(N, A)$, where N represents the set of nodes (locations, charging stations) and A the set of road segments. Unlike traditional logistics models that minimize total fleet distance, this individual-centric model optimizes a weighted objective function Z where:

$$Z = w_1 \cdot T_{\text{total}} + w_2 \cdot C_{\text{total}} + w_3 \cdot P_{\text{penalty}}$$

Where:

- T_{total}: Total travel time, including driving and waiting times at stations.
- C_{total}: Total financial cost, summing energy costs (dynamic pricing) and toll fees.
- $P_{penalty}$: A penalty term for Range Anxiety, derived from the Improved Fitness Function concept found in the literature[3][10], which penalizes arriving at nodes with critically low battery states.
- w_i: Cost/time weights of the fitness function.

The MILP formulation defines binary variables x_{ij} for route selection and continuous variables for arrival time and state-of-charge (SoC). Critical constraints adapted from the *EVRPTW-CRS* (Capacitated Recharging Stations) model include flow and battery conservation, partial recharging strategy and capacitated station logic.[2]

The weights (w_i) in the objective function Z represent the trade-offs between conflicting user goals. The project will allow users to adjust these preferences with a priority setting feature.

2.2. Hybrid Algorithmic Framework

Solving the MILP directly is computationally expensive for long-distance routes. Therefore, the project utilizes a Matheuristic approach, specifically an ALNS metaheuristic that calls a simplified MILP sub-problem to check feasibility and schedule charging stops efficiently.

2.2.1. Adaptive Large Neighborhood Search (ALNS)

The core engine iteratively improves a solution by removing parts of the route and repairing (re-inserting) them.

- Destroy Operators: We employ Random Removal, Greedy Removal to remove the nodes that contribute most to cost, and Cluster (Bomb) Removal for spatial clustering removal. [2]
- Repair Operators: Nodes are re-inserted using Greedy Insertion to get the best immediate cost and k-Regret Insertion to avoid future high costs. [2]
- Acceptance Criterion: A Simulated Annealing (SA) mechanism is used to accept worse solutions occasionally to escape local optima, controlled by a cooling schedule.[2]

EVCRP problem involves strict feasibility constraints. Population based methods like Genetic Algorithm often produce infeasible offspring after crossover operations, requiring complex repair mechanisms. ALNS maintains feasibility more naturally through its "Destroy and Repair" operators. ALNS also allows for the integration of specific problem knowledge through custom operators (like Station-Removal), making it highly effective for the "Partial Recharging" and "Capacitated Station" logic adapted from Bruglieri et al. (2025)[2].

2.2.2. Feasibility Check (The "Matheuristic" Component)

When the ALNS constructs a route, a light-weight MILP model is triggered to optimize the specific *charging schedule* at the chosen stations. This ensures that the arrival times align with station availability (avoiding occupied time slots), ensuring the route is not just theoretically short, but practically feasible.

2.3. Data Integration and Dynamic "Fitness"

To validate the model, the static graph is enriched with dynamic data, transforming standard distance weights d_{ij} into dynamic cost/time weights $w_{ij}(t)$:

- Traffic & Weather: Real-time data from TomTom and Open-Meteo modifies the energy consumption rate r, making it a function of speed and temperature (e.g., battery drain increases in cold weather) rather than a fixed constant.

- Infrastructure: The EPDK Şarj@TR dataset provides station locations (N_{chargers}) and plug types, which defines the charging rate g in the model (Linear or Piecewise-Linear charging profiles).

2.4. Validation and Analysis

The effectiveness of the proposed methodology will be evaluated by comparing the Multi-Criteria ALNS results against a fundamental baseline Dijkstra Shortest Path algorithm. Key performance indicators (KPIs) will include:

- Gap Analysis: Measuring the deviation of the ALNS solution from the optimal MILP solution obtained by a commercial solver on small instances with a small benchmark.
- Cost/Time Savings: Quantifying the reduction in travel cost and time achieved by using partial recharging vs. full recharging strategies and comparing our results with previous studies.[2]
- Reliability Score: A measure of how often the proposed route avoids predicted possible station queues compared to today's navigation.

3 PROJECT MANAGEMENT

3.1 Work Package Timetable

WORK PACKAGE TIMETABLE

WP No	Work Package Name	Start-End Date (Month)	Contributors
1	Literature Review & Requirement Analysis	1 - 3 Month	Ata Türkyılmaz – Yusuf Celal Zeybek
2	Mathematical Modeling (MILP Formulation)	3 - 5 Month	Ata Türkyılmaz
3	Development of Hybrid Algorithmic Framework	5 - 7 Month	Ata Türkyılmaz – Yusuf Celal Zeybek
4	Real-Time Data & Dynamic Fitness Function Integration	6 Month	Ata Türkyılmaz – Yusuf Celal Zeybek
5	Testing and Validation	7 Month	Ata Türkyılmaz – Yusuf Celal Zeybek
6	Prototype Implementation (GUI)	7 - 8 Month	Yusuf Celal Zeybek
7	Documentation	7 - 8 Month	Ata Türkyılmaz – Yusuf Celal Zeybek

3.2 Work Packages

WORK PACKAGE #1	
WP Name: Literature Review & Requirement Analysis	
WP Objective: To establish the theoretical foundation of the project for individual-based EVRP and to define the specific functional requirements for the application.	
WP Start/End Date: 10/2025 - 12/2025 (3 Months)	
WP Tasks **1.1** Conduct a comprehensive literature survey on EVRP **1.2** Conduct a comprehensive literature survey on MILP and ALNS **1.3** Analyze the specific differences between Fleet Optimization and Individual Vehicle Optimization to finalize the problem scope. **1.4** Define the requirements list	**Responsible People** **1.1** Yusuf Celal Zeybek **1.2** Ata Türkyılmaz **1.3** Ata Türkyılmaz **1.4** Yusuf Celal Zeybek
Success Criteria: 1. Literature survey report completed 2. Project well defined in Graduation Project Term Report	

WORK PACKAGE #2	
WP Name: Mathematical Modeling	
WP Objective: To mathematically define the multi-objective optimization problem and create a model to validate heuristic results.	
WP Start/End Date: 12/2025 - 03/2026 (3 Months)	
WP Tasks **2.1** Define the decision variables, sets, and the multi-objective function. **2.2** Formulate specific constraints for partial recharging and capacitated station logic adapted for single vehicles. **2.3** Implement and validate the MILP model using a commercial solver	**Responsible People** **2.1** Ata Türkyılmaz **2.2** Ata Türkyılmaz **2.3** Ata Türkyılmaz
Success Criteria: 1. Small benchmark instances solved correctly.	

WORK PACKAGE #3	
WP Name: Development of Hybrid Algorithmic Framework	
WP Objective: To develop the core optimization engine.	
WP Start/End Date: 02/2026 - 05/2026 (3 Months)	
WP Tasks **3.1** Implement the Adaptive Large Neighborhood Search **3.2** Develop specific Destroy-Repair operators. **3.3** Develop the MILP sub-problem and integrate. **3.4** Test the algorithm	**Responsible People** **3.1** Ata Türkyılmaz - Yusuf Celal Zeybek **3.2** Yusuf Celal Zeybek **3.3** Ata Türkyılmaz **3.4** Yusuf Celal Zeybek
Success Criteria: 1. Fully functional ALNS code implemented 2. Algorithm generates feasible routes for large amount of node instances in under reasonable time	

WORK PACKAGE #4	
WP Name: Real-Time Data & Dynamic Fitness Function Integration	
WP Objective: To integrate the theoretical model into a practical tool by integrating real-world dynamic data	
WP Start/End Date: 03/2026 - 04/2026 (1 Month)	
WP Tasks **4.1** Implement API connections. **4.2** Create the database structure for storing and querying dynamic data. **4.3** Process the dataset to create the charging station graph network. **4.4** Develop the dynamic graph weighing algorithm logic that updates energy consumption based on weather/speed.	**Responsible People** **4.1** Yusuf Celal Zeybek **4.2** Ata Türkyılmaz **4.3** Ata Türkyılmaz - Yusuf Celal Zeybek **4.4** Ata Türkyılmaz - Yusuf Celal Zeybek
Success Criteria: 1. All APIs are integrated 2. Range and time estimation alter dynamically according to changing external factor inputs.	

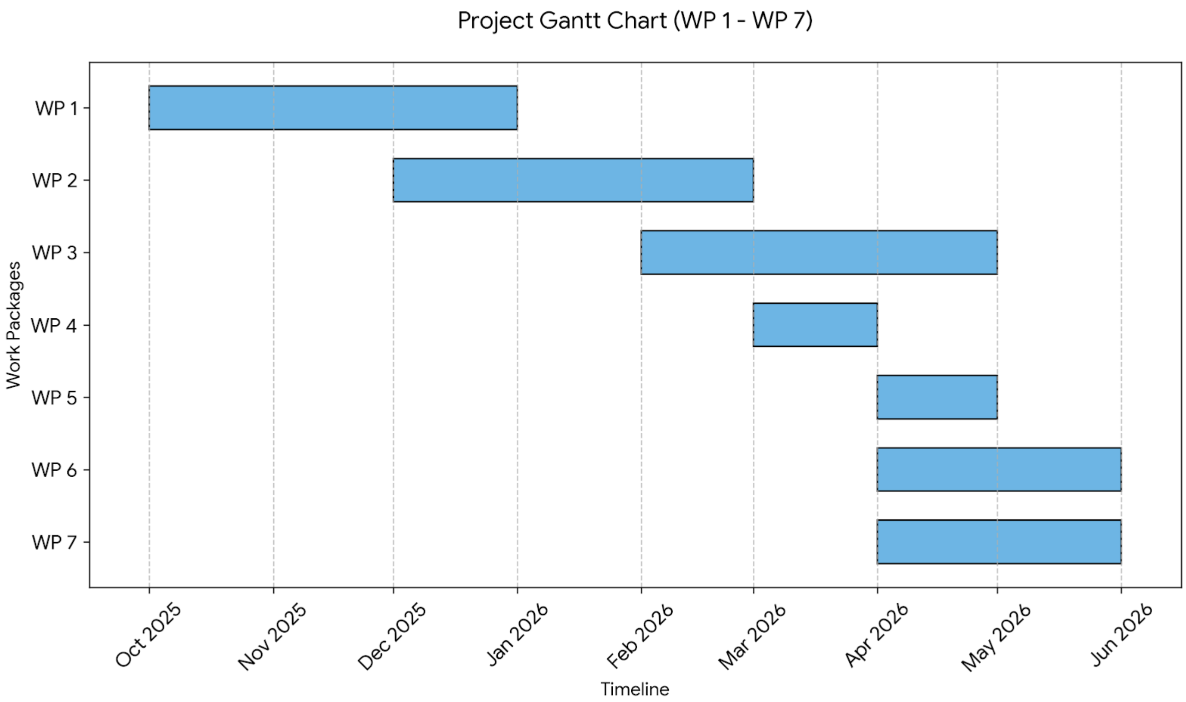
WORK PACKAGE #5	
WP Name: Testing and Validation	
WP Objective: To scientifically validate the proposed solution's effectiveness against baselines.	
WP Start/End Date: 04/2026 - 05/2026 (1 Month)	
WP Tasks **5.1** Perform “Gap Analysis” comparing ALNS results with MILP on small data and Dijkstra on large data. **5.2** Perform “Sensitivity Analysis” to measure the impact of range anxiety and weather on route	**Responsible People** **5.1** Ata Türkyılmaz - Yusuf Celal Zeybek **5.2** Ata Türkyılmaz - Yusuf Celal Zeybek **5.3** Ata Türkyılmaz - Yusuf Celal Zeybek **5.4** Ata Türkyılmaz - Yusuf Celal Zeybek

selection. **5.3** Calculate final KPIs **5.4** Define the requirements list	
Success Criteria: 1. Comparative analysis shows considerable improvement over the baseline	

WORK PACKAGE #6	
WP Name: Prototype Implementation	
WP Objective: To create a user interface that allows common users to interact with the optimization engine and visualize results.	
WP Start/End Date: 04/2026 - 06/2026 (2 Months)	
WP Tasks **6.1** Design the User Interface (UI) layout. **6.2** Connect the frontend to the backend **6.3** Design and implement map visualization components to show routes, charging stops, and waiting times. **6.4** Perform usability testing and bug fixing. **6.5** Gather feedback from users.	**Responsible People** **6.1** Yusuf Celal Zeybek **6.2** Yusuf Celal Zeybek **6.3** Yusuf Celal Zeybek **6.4** Yusuf Celal Zeybek **6.5** Yusuf Celal Zeybek
Success Criteria: 1. Functional prototype deployed. 2. Positive user feedback received.	

WORK PACKAGE #7	
WP Name: Documentation	
WP Objective: To create a final report to record and reveal latest findings	
WP Start/End Date: 04/2026 - 06/2026 (2 Months)	
WP Tasks **7.1** Create the abstract. **7.2** Draw the necessary diagrams **7.3** Identify the references **7.4** Put the report together. **7.5** Find and correct grammar and logic errors.	**Responsible People** **7.1** Ata Türkyılmaz - Yusuf Celal Zeybek **7.2** Yusuf Celal Zeybek **7.3** Ata Türkyılmaz **7.4** Ata Türkyılmaz - Yusuf Celal Zeybek **7.5** Ata Türkyılmaz - Yusuf Celal Zeybek
Success Criteria: 1. Functional prototype deployed. 2. Positive user feedback received.	

3.3 Work Package Gantt Chart



3.4 Risk Analysis

RISK MANAGEMENT TABLE

WP No	Risk Definition	Precautions (Plan B)
1	Inadequate Resources	1. Research will continue with a larger scope such as VRP.
2	Computational Intractability of MILP	1. Linearization and Relaxation 2. Approximation
3	ALNS Runtime Latency	1. Heuristic Feasibility Check
4	API Limitations	1. Explore alternative APIs
4	Data unavailability	1. Static Data Fallback (Offline Mode)
5	Insufficient Improvement	1. Scenario-Specific Stress Testing (Edge Cases)

4. PROJECT OUTCOMES

Output Type	Output	Expected Time of Output
Scientific/Academic (Conference Paper, Journal Paper, Book Chapter etc.)	1. Final Report 2. Conference Paper	1. 8 Months 2. 7 Months

Economical/Commercial/ Social (Product, Prototype, Patent etc.)	1. Prototype	1. 8 Months

5. ALIGNMENT WITH UN SUSTAINABLE DEVELOPMENT GOALS

Impact Area	Does project impact? Direct or indirect?	Impact Definition
Social How the project affects people and society (Who benefits from the project? Does it improve accessibility, inclusion, or equality? Does it reduce manual effort or improve quality of life?) SDG 10 - Reduced Inequalities	Direct & Indirect	The project allows any electric vehicle driver to access advanced routing optimization, previously accessible mainly to large logistics fleets. This directly improves accessibility to sustainable technology for the common people, reducing the technological gap.
Health How the project affects health directly or indirectly (Does it reduce stress, workload, or human error? Does it support health monitoring, safety, or wellbeing? Does it prevent unsafe working conditions?) SDG 3 - Good Health and Well-being	Direct & Indirect	By simplifying the complexity of EV charging, project encourages a broader demographic to adopt electric vehicles.
Security How security is achieved in the project, or how the project increases the security (How is user data protected? What security risks exist?) SDG 16 - Peace, Justice, and Strong Institutions	Direct	The project is designed to process user location and battery data securely by utilizing trusted APIs.
Economic How the project contributes to the economy (Does it reduce costs? Does it improve productivity?) SDG 8 - Decent Work and Economic Growth	Direct & Indirect	The multi-criteria algorithm optimizes not just for time but also for "Total Financial Cost," helping individual users save money on energy directly. By solving key infrastructure usability problems, the project supports the growth of the electric vehicle and charging station sectors indirectly.

Sustainability and Environment How the project contributes to the sustainability and affects the environment (Does it reduce paper usage? Does it optimize energy or resource consumption? Is computer system usage efficient?) SDG 11 - Sustainable cities and communities SDG 13 - Climate Action SDG 12 - Responsible Consumption and Production	Direct	The algorithm uses "Partial Recharging" strategy and weather-aware routing to ensure that battery energy is consumed in the most efficient way possible. The project also facilitates the use of environmentally friendly electric vehicles, making them more widespread and contributing to a significant reduction in carbon footprint.

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